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Materials Design for Compute Thermal Management From Dielectric Fluids to Gallium-Based Cooling Systems

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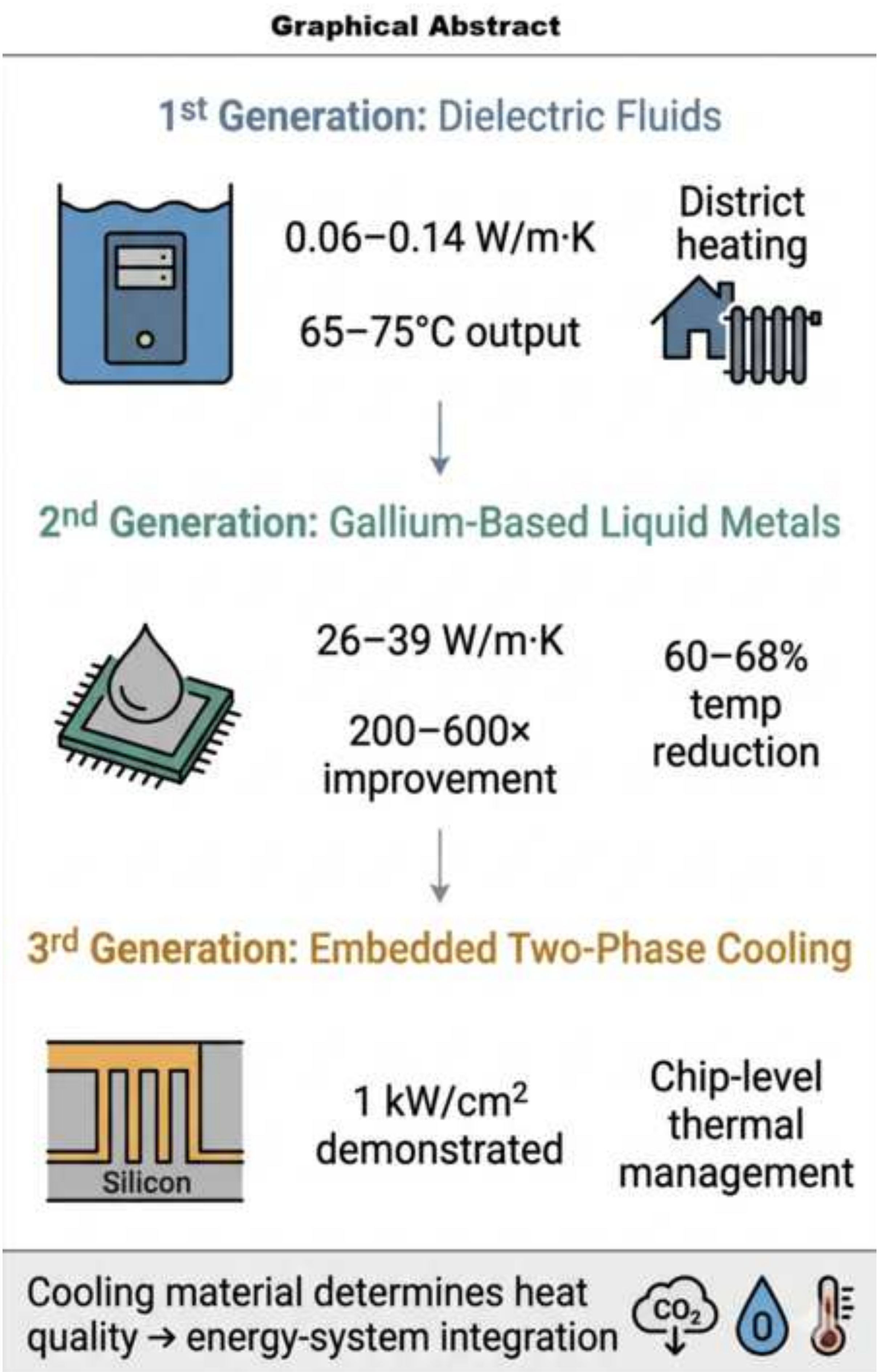
Graphical Abstract
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200–600× improvement
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1 kW/cm²
demonstrated
Chip-level thermal management
Cooling material determines heat quality → energy-system integration
CO₂
0

Materials Design for Compute Thermal Management: From Dielectric Fluids to Gallium-Based Cooling Systems

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Abstract

Nearly all electricity consumed by compute infrastructure exits the facility as thermal energy. Bitcoin mining consumes 138–211 TWh annually; AI data centers are projected to exceed 1,000 TWh by end of 2026. This article examines compute thermal management as a materials design problem. Current first-generation immersion cooling uses dielectric fluids (thermal conductivity 0.06–0.14 W/m·K) that capture waste heat at 65–75°C, directly usable for district heating without heat pumps. Six operational deployments are documented, with lifecycle CO_2 displacement estimated at 2,000–3,800 tonnes per 1.7 MW container per year at 70–85% thermal utilization. Second-generation gallium-based liquid metal systems achieve thermal conductivities of 26–39 W/m·K at the thermal interface, a 200–600× improvement, with experimentally demonstrated 60–68% temperature reduction under pulsed thermal loads. Third-generation embedded two-phase microfluidic cooling, demonstrated under DARPA's ICECool program at 1 kW/cm², points toward chip-level thermal management at heat flux densities that facility-level immersion cannot reach. Each generation of cooling material determines the temperature, quality, and reusability of the waste heat produced. The materials choice is therefore not merely a cooling decision but an energy-systems design decision with consequences for water consumption and carbon displacement.

Keywords – Thermal management; Materials design; Immersion cooling; Gallium; Liquid metals; Phase-change materials; Dielectric fluids; Waste heat recovery; Data centers; District heating

1. Introduction: The Thermal Problem as a Materials Problem

In January 2026, a shipping container arrived in a small town in Scandinavia. Inside were 160 Bitcoin mining rigs submerged in dielectric fluid, generating 1.5 MW of thermal energy that was pumped into the town's district heating grid, warming approximately 2,000 homes (assuming ~5 MWh/home/year heating demand). The company that built it, Latvia-based Power Mining, described itself as a heating company that happens to mine Bitcoin [5]. Simultaneously, a greenhouse in Manitoba was being heated by 360 liquid-cooled mining servers capturing approximately 90% of consumed electricity as heat at 75°C [6]. And in Dublin, students at the Technological University attended lectures in buildings heated by waste heat from an Amazon Web Services data center [15].

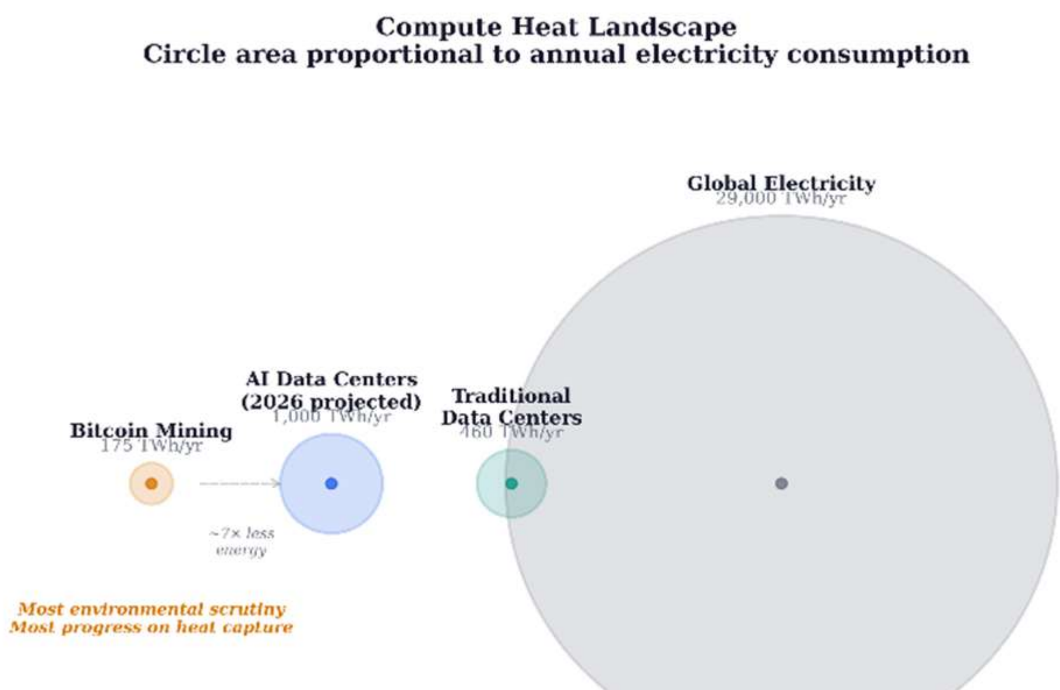
These deployments share a common engineering feature: the cooling material determines the output temperature, and the output temperature determines the heat's industrial utility. Air-cooled systems reject heat at 25–35°C—too low for district heating networks, which require 60–80°C supply temperatures [4]. Immersion in dielectric fluid raises the capture temperature to 65–75°C, crossing the district heating threshold. Gallium-based liquid metals, with thermal conductivities two to three orders of magnitude higher than dielectric fluids, could raise output temperatures above 80°C while simultaneously managing the escalating heat flux densities of AI accelerators. The choice of cooling material is therefore an energy-systems design decision, not merely a thermal engineering one.

The data center waste heat recovery literature has grown substantially in recent years. Huang et al. [33] reviewed data centers as energy prosumers in district heating systems; Wahlroos et al. [34] analyzed waste heat utilization in Nordic data centers; Yuan et al. [35, 36] published comprehensive reviews of data center waste heat recovery technologies and their integration with district heating networks. These studies approach the problem from the system and network perspective. The present article complements this literature by approaching the problem from the cooling *material* perspective: examining how the thermal properties of the cooling medium itself determine the temperature, quality, and reusability of waste heat, and tracing a three-generation materials progression that the existing literature has not framed.

This article examines three generations of compute cooling materials—dielectric fluids, gallium-based liquid metals, and embedded two-phase microfluidic systems—through the lens of materials design. It documents operational deployments that validate first-generation technology, reviews the materials science frontier of second- and third-generation systems, and identifies the engineering challenges that separate laboratory demonstrations from facility-scale deployment.

The energy baseline is drawn from the Cambridge Centre for Alternative Finance's Digital Mining Industry Report (April 2025), which surveyed 49 firms covering 48% of global hash rate [1]: Bitcoin mining consumes 138 TWh/yr (upper estimate 211 TWh [2]) with a sustainable energy share of 52.4% (42.6% renewables, 9.8% nuclear). AI data centers are projected by the IEA to exceed 1,000 TWh by end of 2026, more than 7× Bitcoin mining's consumption [3].

Selection bias note: The Cambridge survey's respondents are disproportionately publicly listed (41%) and North American. The unreported 52% includes operations in Central Asia and Africa where coal and gas typically dominate. The true global sustainable share may be materially lower than 52.4% [1].



[Figure 1] *Compute heat landscape. Circle area is proportional to annual electricity consumption. Bitcoin mining receives the most environmental scrutiny while consuming the least electricity and having made the most progress on* *heat capture.*

2. Thermodynamic Framework

Every processor converts electricity into computation and heat at a ratio of approximately 100%. The first law of thermodynamics guarantees energy conservation; the second law constrains its quality. Electricity entering a facility is high-grade energy; heat exiting at 65°C is low-grade, suitable for space and water heating but not for mechanical or electronic work.

A heat pump delivers 3–4 joules of thermal energy per joule of electricity consumed (COP of 3–4). A mining rig delivers approximately 1 joule of heat per joule of electricity, plus cryptocurrency revenue. Mining-as-heating is therefore thermodynamically inferior to a heat pump for pure heating purposes and is economically competitive only when computation revenue offsets the COP gap. The materials question is how to maximize the temperature and quality of the rejected heat so that the gap narrows and the co-product value increases.

3. First-Generation Cooling Materials: Dielectric Fluids

3.1. Thermal properties and performance envelope

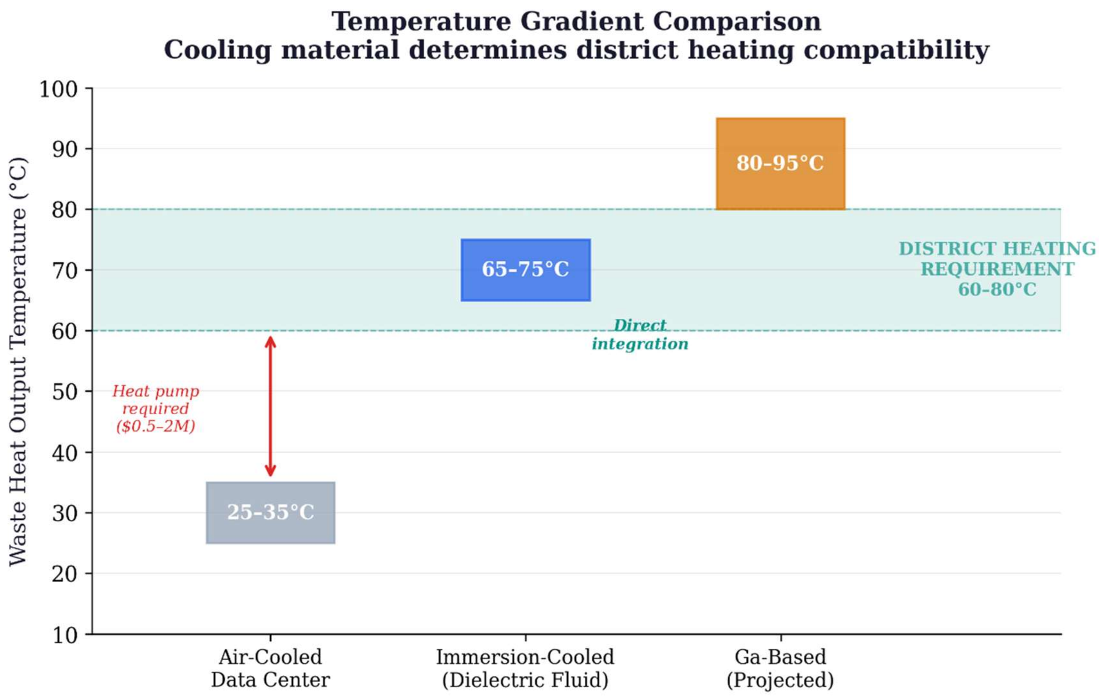
Single-phase immersion fluids—engineered hydrocarbons or synthetic esters—offer thermal conductivities of 0.06–0.14 W/m·K, specific heat capacities of 1.5–2.1 kJ/kg·K, and operating ranges from –40°C to +200°C [9, 10, 11]. They are chemically inert, electrically non-conductive, and non-flammable. Two-phase fluorocarbon fluids offer higher heat transfer coefficients through phase-change latent heat, but several formulations fall under PFAS classification; the EU’s proposed PFAS restriction under REACH may limit future availability [12, 13]. The broader liquid cooling market for AI data centers is projected to grow from $2.8 billion in 2025 to over $21 billion by 2032.

The EU Energy Efficiency Directive (2023/1791) now mandates waste heat recovery for new data centers above 1 MW within the European Union [13]. This regulatory shift changes the question from whether data centers should recover heat to *how* they should recover it—and the “how” is precisely where the materials framework developed in this article becomes policy-relevant.

Single-phase fluids exhibit measurable total acid number (TAN) increases after 8,000–12,000 hours of continuous operation above 60°C, indicating oxidative degradation. Replacement intervals of 5–8 years are typical for closed-loop systems with adequate oxygen exclusion [9, 10, 11].

3.2. The temperature gradient breakthrough

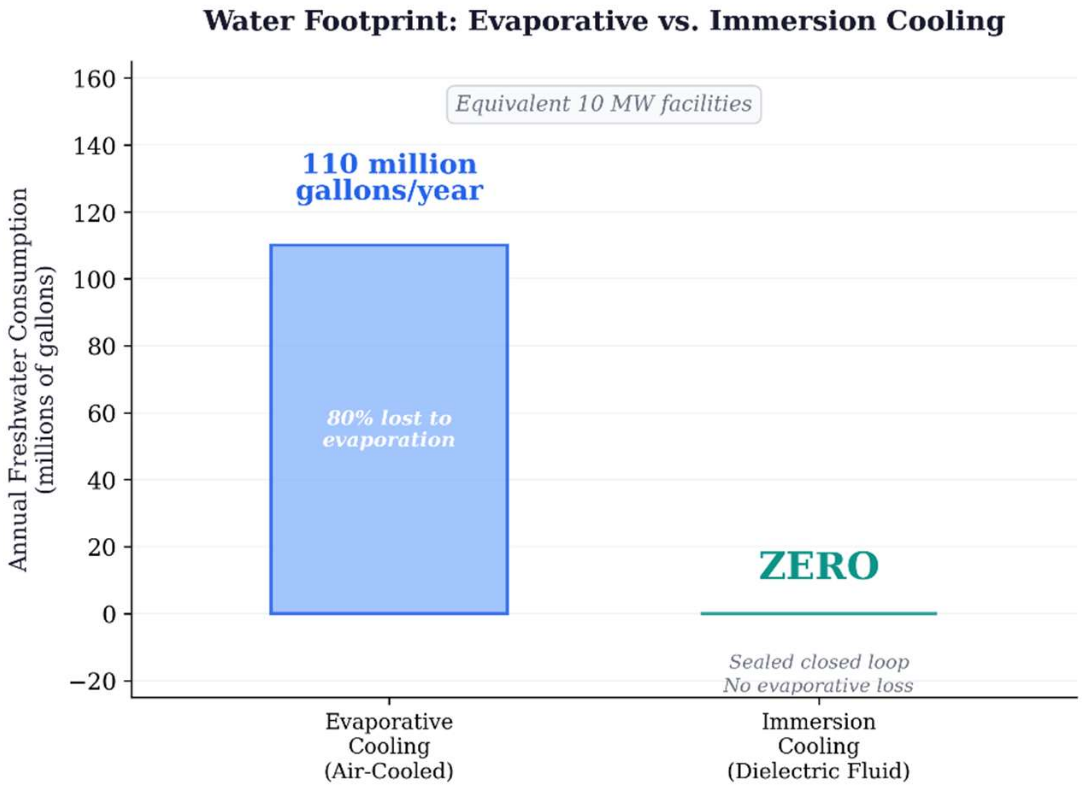
District heating networks require supply temperatures of 60–80°C [4]. Air-cooled data centers produce waste heat at 25–35°C, below this threshold. Bridging the gap requires heat pumps ($500K–$2M additional capital). Immersion cooling changes this fundamentally. Submerging hardware in dielectric fluid captures heat at 65–75°C—directly within the district heating range. No heat pump is required. Power Mining’s containers produce heat at 65°C; Canaan’s system produces water at 75°C [5, 6]. This is the article’s central materials-design insight: cooling material selection determines not only thermal performance but downstream energy-system integration.



[Figure 2] *Temperature gradient comparison. Air-cooled facilities (25–35°C output) fall below district heating requirements (60–80°C). Immersion-cooled systems eliminate the gap, enabling direct grid integration without heat pumps.*

3.3. Water elimination

Evaporative cooling consumes approximately 110 million gallons of freshwater annually for a medium facility (~10 MW), with 80% permanently lost through evaporation [7]. For an equivalent 10 MW immersion-cooled facility, water consumption is zero—dielectric fluid circulates in a sealed closed loop with no evaporative loss. Each 100-word AI prompt processed by an evaporatively cooled data center consumes approximately 519 mL of water [8]. Immersion cooling represents a complete elimination of freshwater demand from the cooling system. This claim refers to the cooling loop itself; the water footprint of electricity generation that powers the compute is a separate lifecycle consideration.



[Figure 3] *Water footprint comparison for equivalent 10 MW facilities. Evaporative cooling: ~110 million gallons/year; immersion cooling: zero.*

4. Engineering Validation: Operational Deployments

4.1. Documented heat-reuse systems

Six operational deployments validate the first-generation materials approach. These are illustrative examples within a broader trend; Yuan et al. [35] documented over 20 operational or planned data center heat-reuse installations across Europe. The Dublin AWS scheme is the only deployment with independently measured emissions data: 704 tCO_2 abated in 2024, verified by TU Dublin [15]. For Bitcoin mining deployments, environmental claims are based on engineering projections from reported specifications, not measured outcomes.

Documented Heat-Reuse Deployments
Gray italic = unverified engineering projections; green = independently measured

Deployment	Location	Thermal Output	Output Temp.	Application	CO_2 Data Verified?
Power Mining	Scandinavia	1.7 MW	65°C	District heating (~2,000 homes)	*No — eng. projection*
Canaan/Bitforest	Manitoba, CA	~0.5 MW	75°C	Greenhouse heating	*No — eng. projection*
MintGreen	Lonsdale, BC	~0.3 MW	~65°C	Commercial building (96% of load)	*No — eng. projection*
MARA Digital	Finland	~2 MW	~65°C	District heating	*No — eng. projection*
AWS Dublin	Dublin, IE	N/A	N/A	University campus	Yes — 704 tCO_2 (2024)
Microsoft	Helsinki, FI	N/A	N/A	District heating (planned)	*No — planned*

[Figure 4] *Standardized comparison of six heat-reuse deployments. Gray italic = unverified engineering projections; green = independently measured. Dublin AWS is the only deployment with third-party CO_2 measurement.*

4.2. Lifecycle CO_2 displacement model

The following model estimates annual CO_2 displacement for a single Power Mining container (1.7 MW thermal output, 65°C, operating in a Nordic climate with 8-month heating season). Thermal output of 1.7 MW continuous over approximately 5,840 annual operating hours yields 9,928 MWh of delivered thermal energy. In practice, thermal utilization rates of 70–85% are typical for well-integrated district heating connections due to demand fluctuation, maintenance downtime, and network constraints, reducing effective delivery to approximately 6,950–8,440 MWh. Depending on the displaced fuel—natural gas (0.2 tCO_2/MWh), fuel oil (0.27 tCO_2/MWh), or peat (0.38 tCO_2/MWh)—annual displacement ranges from approximately 1,400 to 3,200 tonnes of CO_2 per container after accounting for utilization losses. The net climate benefit is the difference between displaced heating emissions and mining electricity emissions (~0.29 tCO_2/MWh at 52.4% sustainable mix).

4.3. Scalability and geographic constraints

Finland, Sweden, Denmark, and Norway deliver more than 50% of heating through district networks. Compute heat integration works in these regions because of cold climate (8–10 months demand), existing pipe infrastructure, supportive municipal policy, and high conventional energy costs [4]. Many Finnish district heating systems already source heat from co-generation plants; in these systems, the marginal climate benefit of mining heat is reduced because the displaced source was already efficient [37]. The strongest case exists where networks depend on standalone fossil boilers.

District heating networks are designed for 30-year lifespans. Mining hardware is obsolete in 3–5 years; data center hardware in 7–10. Municipalities evaluating mining-heated districts must plan for multiple hardware refresh cycles per infrastructure lifecycle [18].

It is worth noting that Bitcoin mining also provides rapid demand response that no other large-scale compute workload can match. Riot Platforms' 700 MW Texas facility curtailed more than 95% of power usage during the August 2023 heat wave, earning $31.7M in ERCOT curtailment credits in a single month [S1, S2]. ERCOT large-load requests reached 226 GW in 2025, with AI companies now comprising 73% of new applications. The demand response dimension is treated in detail in the Supplementary Material (Section S1).

Table 1. Three-Generation Materials Comparison

Property	1st Gen: Dielectric Fluids	2nd Gen: Ga-Based Liquid Metals	3rd Gen: Embedded Two-Phase
Thermal conductivity	0.06–0.14 W/m·K	26–39 W/m·K (TIM); loop TBD	Fluid-dependent; >1 kW/cm² removal
Output temperature	65–75°C	>80°C (projected)	60–80°C (tunable via pressure)
Electrical conductivity	Non-conductive	~3.5×10■ S/m (hazard)	Non-conductive
Deployment scale	Facility-level (MW)	TIM: consumer; Loop: lab only	Chip-level (demo); facility TBD
TRL	8–9 (commercial)	TIM: 8–9; Loop: 3–4	4–5 (DARPA demo)
Water consumption	Zero (closed loop)	Zero (closed loop)	Zero (closed loop)
Key challenge	Low thermal conductivity; degradation	Al corrosion; electrical containment	Mfg scalability; microchannel reliability
DH compatibility	Direct (no heat pump)	Direct + higher T (projected)	Direct (pressure-tunable)

Table 1. Three-generation materials comparison. TRL = Technology Readiness Level. DH = District Heating. TIM (milligrams, chip-level) and cooling loop (kilograms, facility-level) are distinguished for Ga-based systems.

5. Second-Generation Materials: Gallium-Based Liquid Metal Systems

5.1. Thermal performance

Pure gallium exhibits a thermal conductivity of 33.7 W/m·K in the solid state and 24 W/m·K as a liquid, with a latent heat of 80,300 J/kg at a melting point of 29.8°C [19]. GaInSn eutectic alloys (Galinstan: 68.5% Ga, 21.5% In, 10% Sn) achieve thermal conductivities of 26–39 W/m·K as thermal interface materials, with optimized formulations reaching higher values depending on composition, bond line thickness, surface roughness, and contact pressure [20, 21].

It is essential to distinguish two applications of gallium-based materials with very different engineering requirements and readiness levels. Ga-based *thermal interface materials* (TIMs), applied at the die-to-heat-sink junction in milligram quantities, are commercially mature: Sony’s PlayStation 5 uses a Galinstan-based TIM in mass production, and Indium Corporation’s GalliTHERM product line targets HPC semiconductor applications (TRL 8–9). Ga-based *cooling loops*, circulating kilograms of liquid metal through meters of facility piping, are essentially undemonstrated at scale (TRL 3–4). Bridging this gap is the central scaling challenge for the second generation.

5.2. Phase-change properties and figures of merit

Boteler et al. demonstrated that for a ΔT of 20°C, gallium absorbs 7× more energy per unit volume than copper and 3× more than organic paraffin PCMs [19]. Shao et al. developed a figure-of-merit for PCM cooling capacity; metallic alloys achieve FOMs 1–2 orders of magnitude higher than organic PCMs (2,603 × 10^3 $W^2s/(Km$■) for Bi-based metallic alloy vs. 22 × 10^3 for paraffin wax) [22].

5.3. Experimental validation under pulsed thermal loads

Gonzalez-Nino et al. demonstrated that metallic PCMs reduced temperature rise by 60–68% under 19 ms pulsed thermal loads at heat fluxes up to 338 W/cm², compared to only 14–17% for the best organic PCM [23]. At 160 W pulse power, the metallic PCM maintained temperature within 5°C of its melting point while the dielectric baseline rose 119°C.

Ma and Liu demonstrated thermal conductivity enhancement ratios up to 2.3× with 20% CNT volume fraction in liquid gallium [24]. This result, while promising, has not been reproduced at scale; CNT dispersion in liquid gallium faces sedimentation, agglomeration, and oxide entrapment challenges at volumes beyond the laboratory. The liquid metal TIM market reached $305.9 million in 2025, projected to $479.7 million by 2032 [25].

6. Engineering Challenges for Gallium-Based Cooling Systems

6.1. Corrosion and material compatibility

Gallium alloys aggressively attack aluminum via liquid metal embrittlement. Deng and Liu found that 6063 aluminum alloy suffered complete gallium infiltration within five hours at 120°C [29]. T2 copper exhibited slow corrosion at ~30–60 μm/month at 60°C. 1Cr18Ni9 stainless steel showed zero detectable gallium infiltration after 30 days at 60°C [29]. The practical materials palette: nickel-plated copper or 316L stainless steel throughout.

6.2. Electrical conductivity and containment

Liquid metals are electrically conductive (~3.5 × 10■ S/m). A dielectric fluid leak is a maintenance event; a liquid metal leak is a catastrophic short-circuit risk. Gallium-based systems require sealed, isolated cooling channels rather than open-bath immersion.

6.3. Surface tension and wettability

Gallium exhibits high surface tension (~709 mN/m vs. ~73 mN/m for water), requiring oxide-layer engineering for uniform thermal contact.

6.4. Supercooling mitigation

Bulk gallium can remain liquid to approximately −38°C, nearly 68°C below its 29.8°C melting point. Zhang et al. demonstrated that 0.5 wt.% TeO_2 particles reduced supercooling by 44% [30].

6.5. Toxicity considerations

Gallium itself exhibits low toxicity. However, Galinstan contains ~21.5% indium, and indium compounds have documented pulmonary toxicity (indium lung disease) associated with occupational exposure. At facility scale, handling protocols must address inhalation risks from aerosolized particles during maintenance, leak cleanup, and alloy preparation.

6.6. Implications for heat-reuse systems

A gallium cooling loop could capture waste heat at higher temperatures with lower thermal resistance, potentially enabling output temperatures above 80°C. If Ga-based systems can deliver output temperatures consistently above 80°C, they may eliminate the need for heat pump boosting that some immersion-cooled installations still require for network compatibility. This is a projection, not yet demonstrated in district heating applications.

Boteler et al. demonstrated time-scale matched PCM systems—pairing metallic PCMs with organic PCMs—enabling thermal solutions 25–50% lighter than pure metallic systems [19].

7. Gallium Supply Chain Constraints

Gallium is recovered as a by-product of aluminum and zinc refining [26]. ~95% of consumption goes to GaAs wafer fabrication. China accounts for ~80% of primary production and imposed export controls in July 2023 [27]. A lifecycle analysis of global gallium flows (2000–2021) found that of 5,862 tonnes entering the industrial cycle, only 967 tonnes remain in active use, with 83.5% wasted and 4,446 tonnes in recoverable manufacturing/electronic waste [31].

Facility-scale cooling loops are circulatory systems in which gallium recirculates indefinitely; the operative supply chain metric is make-up volume for leak and maintenance losses, not total fill volume, which substantially reduces ongoing demand relative to the semiconductor pathway where gallium is permanently embedded in wafers. The supply constraint is manageable at current projected adoption rates but must be monitored given GaN power electronics demand growth.

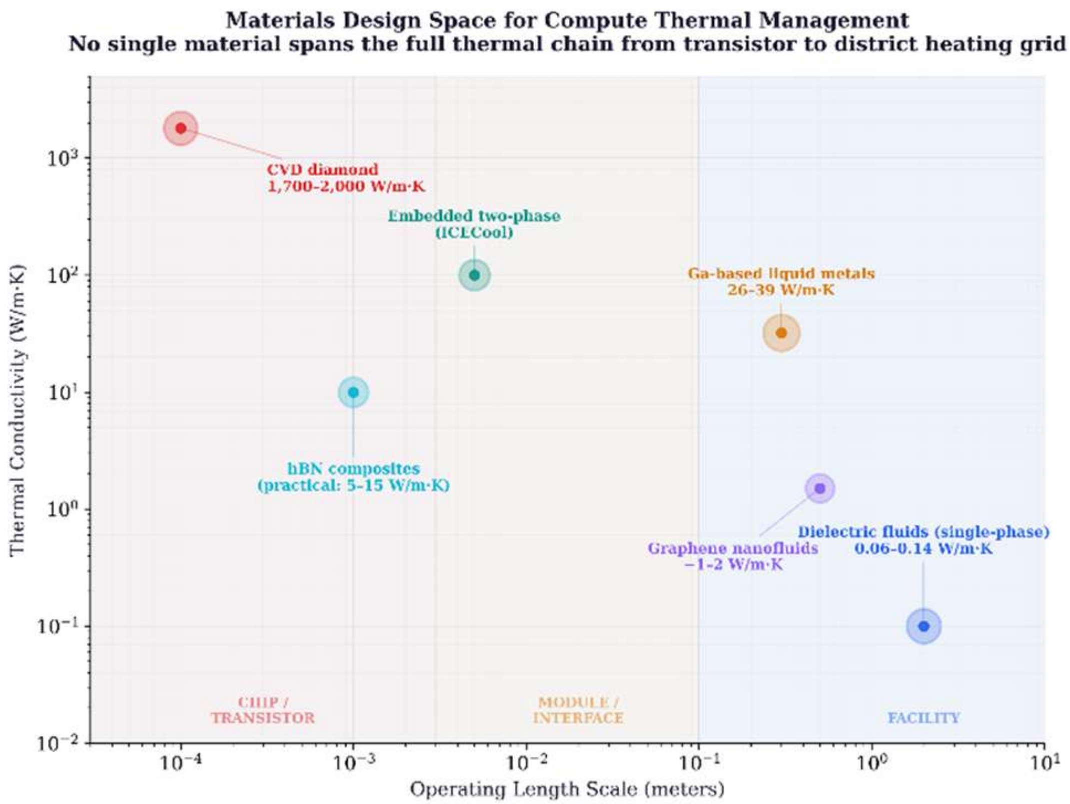
8. Third-Generation Materials: Embedded Two-Phase Cooling

8.1. The ICECool program

DARPA's ICECool program demonstrated chip-level heat flux removal exceeding 1 kW/cm², with hot-spot cooling approaching 5 kW/cm² [28]. IBM demonstrated microchannels in a production microprocessor achieving junction temperatures ~25°C lower than air-cooled baseline while reducing power by >10 W. R1234ze (GWP < 1) can produce exit temperatures in the 60–80°C range suitable for district heating. Microsoft collaborated with Corintis in 2025 on AI-optimized microchannels for Azure Maia.

8.2. Where the other materials fit

CVD diamond (1,700–2,000 W/m·K) manages hotspots at the transistor level; gallium transports heat through meters of facility piping. They are complementary, not competitive. Graphene nanofluids achieve ~1–2 W/m·K—meaningful improvement over dielectrics but 15–30× below gallium. Hexagonal boron nitride offers theoretical parity with diamond plus electrical insulation, but practical composites achieve only 5–15 W/m·K—a decade-scale manufacturing gap.



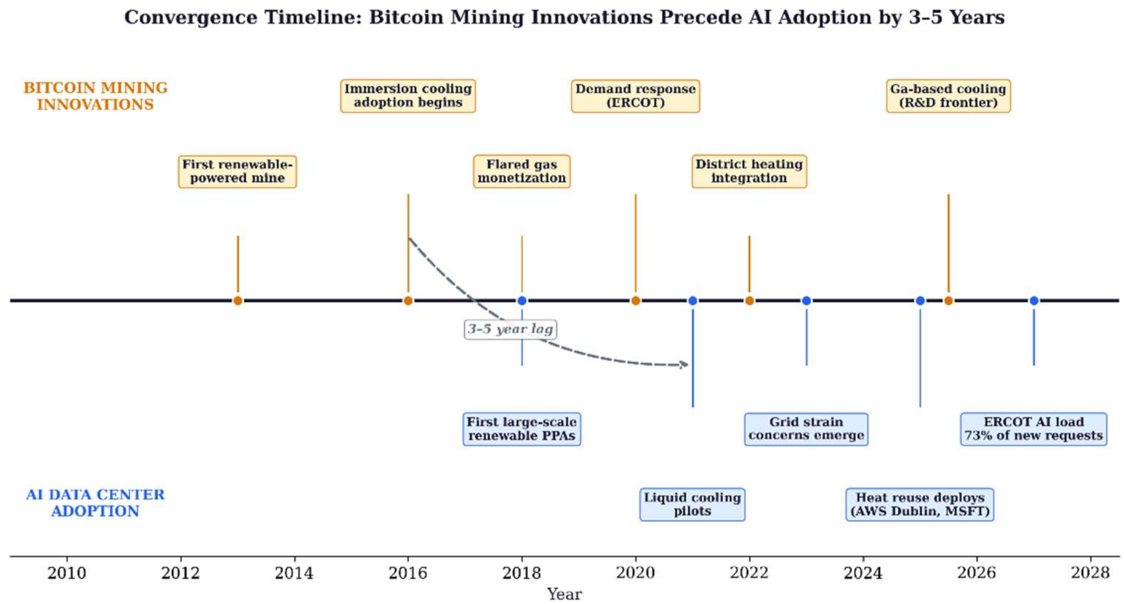
[Figure 5] Three-generation materials comparison. Thermal conductivity (W/m·K) plotted against operating length scale. Each material class occupies a distinct region; no single material spans the full thermal chain from transistor to district heating grid.

9. Conclusion

First-generation dielectric fluids enabled the temperature gradient breakthrough: raising waste heat output from 25–35°C to 65–75°C, crossing the district heating threshold. Six deployments validate this with estimated CO_2 displacement of 1,400–3,200 tonnes per 1.7 MW container per year at realistic utilization rates, though verified emissions data exists for only one deployment. Second-generation gallium-based liquid metals (26–39 W/m·K) offer 200–600× improvement with 60–68% temperature reduction under pulsed loads, but face challenges in corrosion, electrical conductivity, indium toxicity, and supply chain concentration. The distinction between mature TIM applications (TRL 8–9) and undemonstrated facility-scale cooling loops (TRL 3–4) must be clearly maintained. Third-generation embedded two-phase cooling at 1 kW/cm² points toward chip-level management with pressure-tunable exit temperatures for district heating integration.

Heat reuse does not make Bitcoin mining carbon-neutral. The renewable share is 52%, not 100%. ASIC obsolescence generates 15,000–30,000 tonnes of e-waste annually. Heat-reuse benefits apply primarily in cold climates with existing district heating infrastructure.

The sector that received the most environmental scrutiny—Bitcoin mining—has made the most environmental progress, pioneering immersion cooling, demand response, and heat reuse years before the AI industry confronted the same challenges at larger scale. The materials design frontier documented here is the engineering path along which that progress will continue.



[Figure 6] *Convergence timeline. Bitcoin mining thermal innovations (above axis) consistently precede AI data center adoption (below axis) by 3–5 years across cooling materials, power sourcing, and heat reuse.*

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author holds no material positions in Bitcoin, mining companies, or energy firms discussed herein.

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Use of AI-Assisted Tools

The author used Anthropic Claude (Opus) for editorial revision, structural reorganization, image production, and formatting. All analytical content, materials science review, engineering analysis, and conclusions are the author's work.

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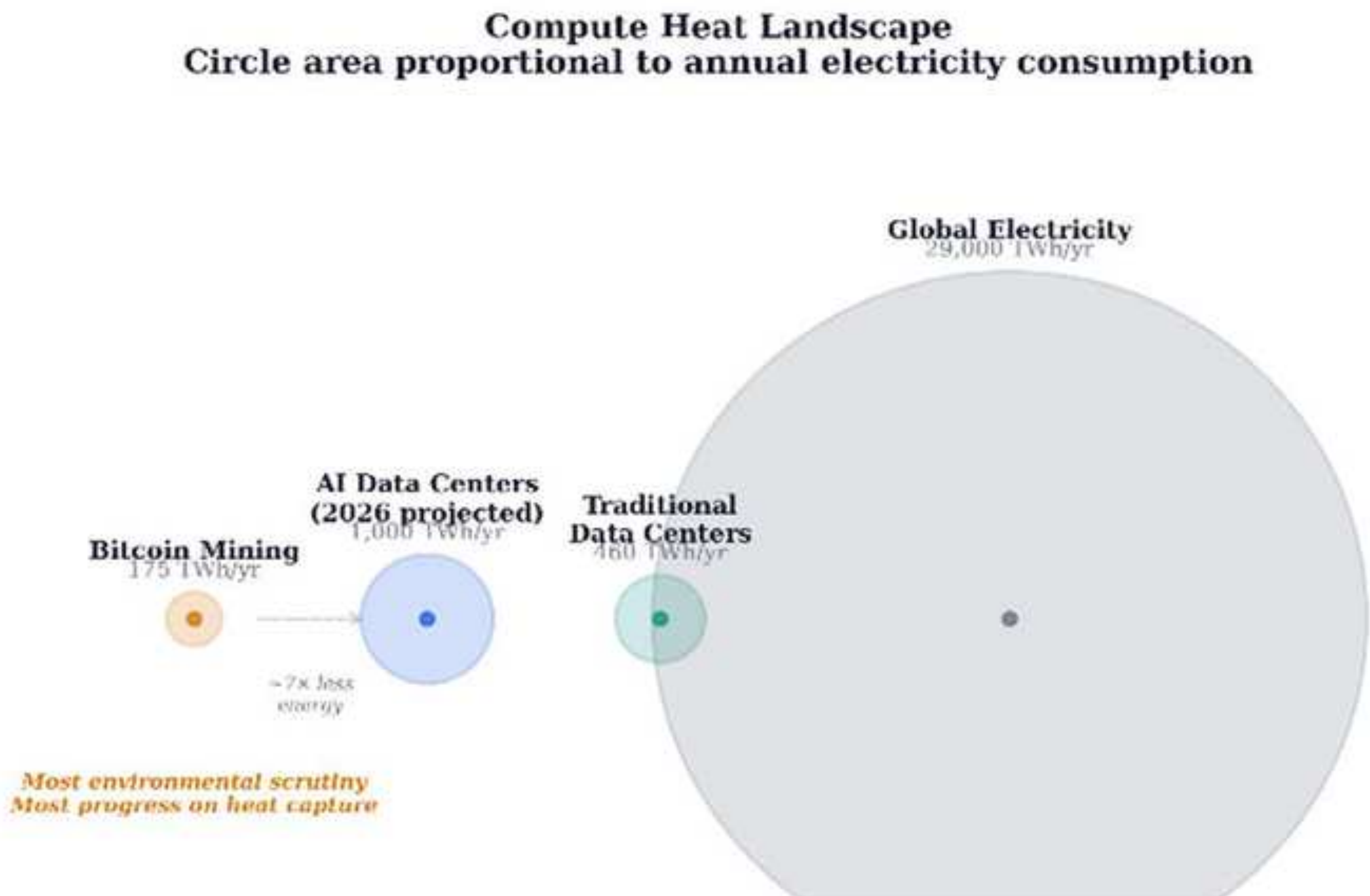
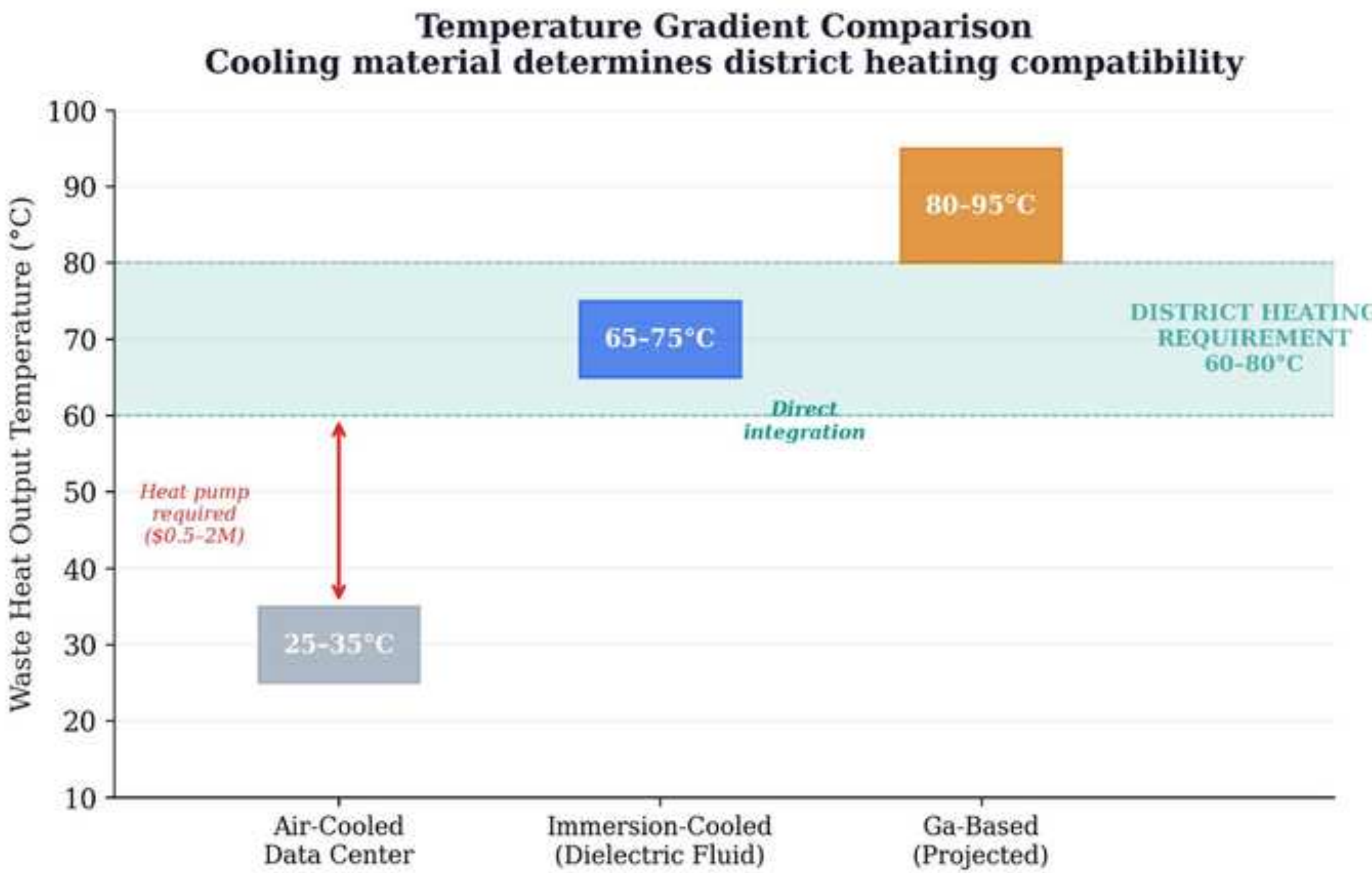
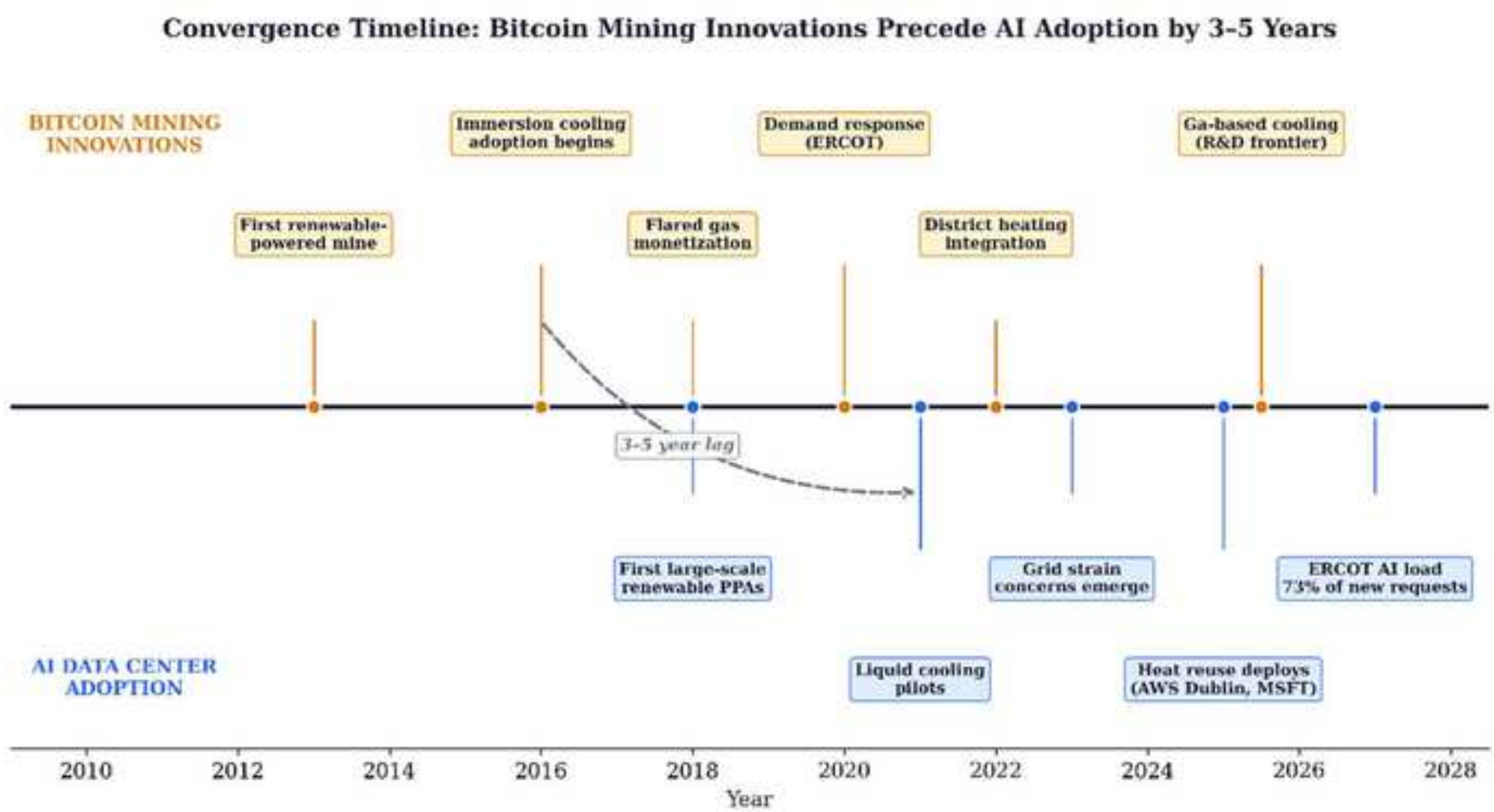
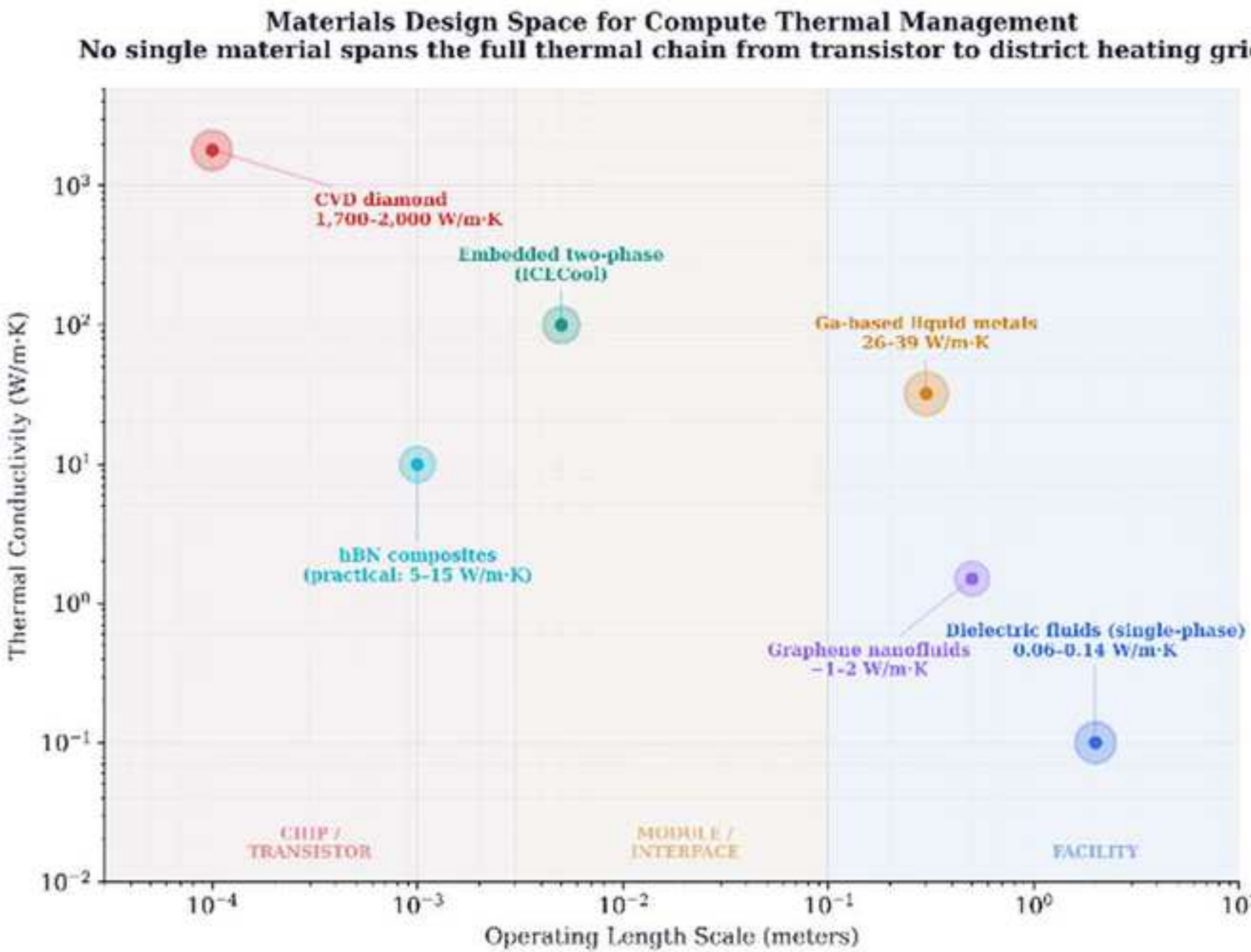
Note on literature scope: The author does not claim the reference set is exhaustive. Readers should consult Deng and Liu, Liquid Metals for Advanced Energy Applications (AIP, 2022) and Advanced Liquid Metal Cooling (Springer, 2024) for additional references.

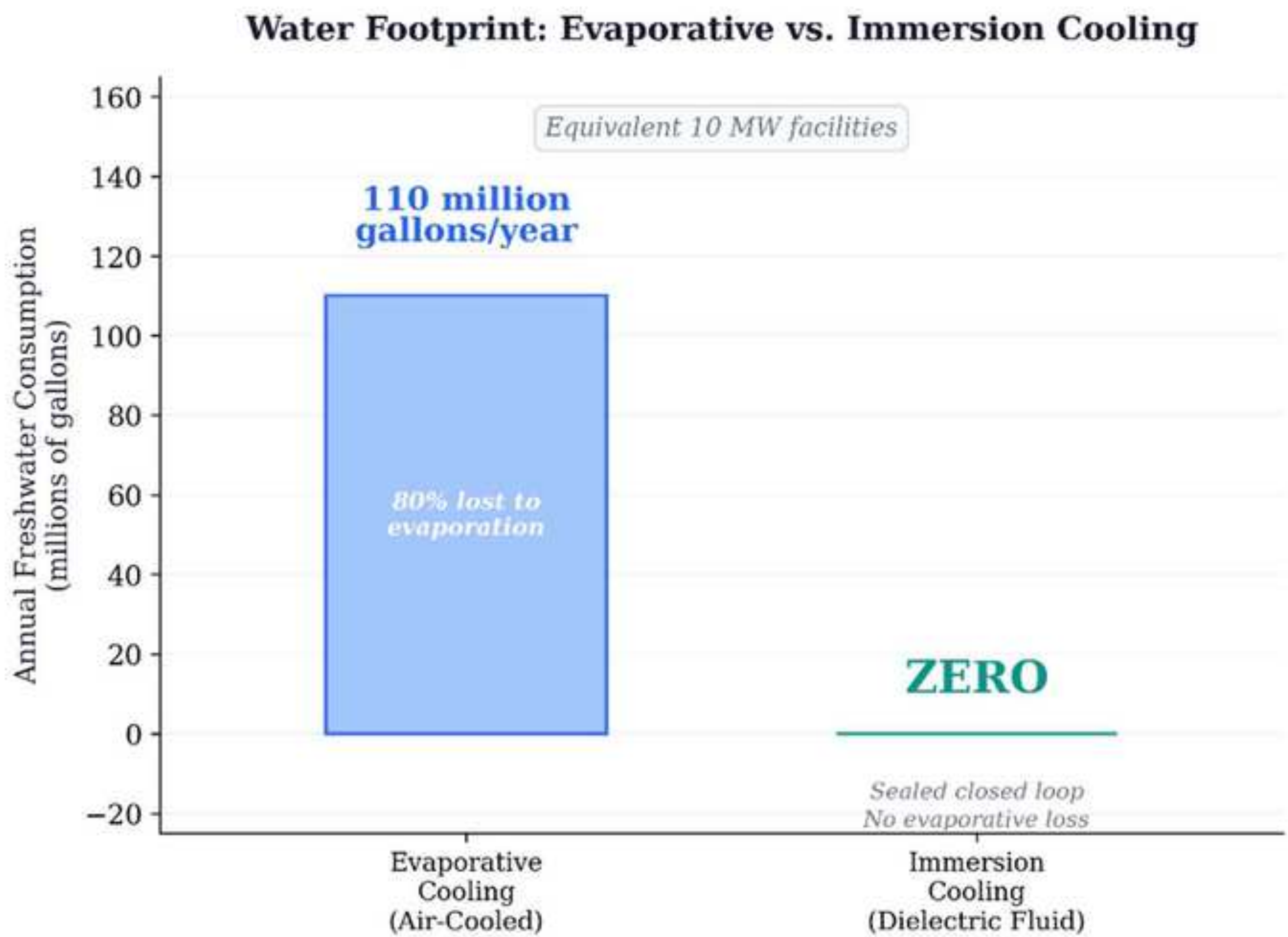

Compute Heat Landscape
Circle area proportional to annual electricity consumption
Global Electricity
29,000 TWh/yr
AI Data Centers
(2026 projected)
1,000 TWh/yr
Traditional
Data Centers
460 TWh/yr
Bitcoin Mining
175 TWh/yr
~7x less
energy
Most environmental scrutiny
Most progress on heat capture


Temperature Gradient Comparison
Cooling material determines district heating compatibility
Waste Heat Output Temperature (°C)
100
90
80
70
60
50
40
30
20
10
80-95°C
65-75°C
25-35°C
DISTRICT HEATING REQUIREMENT 60-80°C
Direct integration
Heat pump required ($0.5-2M)
Air-Cooled Data Center
Immersion-Cooled (Dielectric Fluid)
Ga-Based (Projected)

Click here to access/download;Figure;Fig6_Convergence_Timeline.jpg




Materials Design Space for Compute Thermal Management
No single material spans the full thermal chain from transistor to district heating grid
CVD diamond
1,700-2,000 W/m·K
Embedded two-phase
(ICECool)
Ga-based liquid metals
26-39 W/m·K
hBN composites
(practical: 5-15 W/m·K)
Graphene nanofluids
~1-2 W/m·K
Dielectric fluids (single-phase)
0.06-0.14 W/m·K
CHIP /
TRANSISTOR
MODULE /
INTERFACE
FACILITY
Thermal Conductivity (W/m·K)
Operating Length Scale (meters)
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}


Water Footprint: Evaporative vs. Immersion Cooling
Equivalent 10 MW facilities
110 million gallons/year
80% lost to evaporation
ZERO
Sealed closed loop
No evaporative loss
Annual Freshwater Consumption (millions of gallons)
160
140
120
100
80
60
40
20
0
−20
Evaporative Cooling (Air-Cooled)
Immersion Cooling (Dielectric Fluid)

Documented Heat-Reuse Deployments

Gray italic = unverified engineering projections; green = independently measured

Deployment	Location	Thermal Output	Output Temp.	Application	CO_2 Data Verified?
Power Mining	Scandinavia	1.7 MW	65°C	District heating (~2,000 homes)	*No — eng. projection*
Canaan/Bitforest	Manitoba, CA	~0.5 MW	75°C	Greenhouse heating	*No — eng. projection*
MintGreen	Lonsdale, BC	~0.3 MW	~65°C	Commercial building (96% of load)	*No — eng. projection*
MARA Digital	Finland	~2 MW	~65°C	District heating	*No — eng. projection*
AWS Dublin	Dublin, IE	N/A	N/A	University campus	Yes — 704 tCO_2 (2024)
Microsoft	Helsinki, FI	N/A	N/A	District heating (planned)	*No — planned*

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The author holds no material positions in Bitcoin, mining companies, or energy firms discussed herein.

CRediT Author Statement

Materials Design for Compute Thermal Management: From Dielectric Fluids to Gallium-Based Cooling Systems

Michael Bustamante

Michael Bustamante: Conceptualization, Investigation, Formal analysis, Writing – original draft, Writing – review & editing, Visualization, Project administration.